

Fundamentals of "C" – Language

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Data Type	Size (Typical)	Stores	Format Specifier	Example
char	1 byte	Single character	%c	'A'
int	4 bytes	Whole numbers	%d	25
float	4 bytes	Decimal numbers (6–7 digits)	%f	12.5
double	8 bytes	Decimal numbers (15–16 digits)	%lf	12.567891234
short int	2 bytes	Small integers	%hd	100
long int	4 or 8 bytes	Large integers	%ld	123456789
unsigned int	4 bytes	Positive integers only	%u	500

4. Variables & Constants & Declaring Variables

5. Basic Data Types

6. Program for How to Use Integer

7. Program for How to use Float

8. Program for How to use Double

9. Program for How to use Char

10. Program for How to use String

```
#include<stdio.h>
int main()
{
    int a = 10;
    float b = 5.141598989;
    double e = 3.141598989;
    char c = 'A';
    long int d = 1000000;
    char res[] = "Manikanta kotti";

    printf("%d\n", a);    // Output: 10

    printf("%f\n", b);    // Output: 5.141599
    printf("%.1f\n", b);  // Output: 5.1
    printf("%.1f\n", b);  // Output: 5.141599

    printf("%c\n", c);    // Output: A
    printf("%lf\n", e);    // Output: 3.141599
    printf("%.3lf\n", e);  // Output: 3.142
    printf("%ld\n", d);    // Output: 1000000

    printf("%s\n", res);   // Output: Manikanta kotti

    return 0;
}
```

11.printf() & Scanf()

```
#include <stdio.h>
int main()
{
    int a;
    float b;
    double c;
    char d;
    char e[20];

    printf("enter the int\t");
    scanf("%d",&a);
    printf("a = %d\n",a);    // Output: a = 10

    printf("enter the float\t");
```

```

scanf("%f",&b);
printf("b = %f\n",b);    // Output: b = 5.140000

printf("enter the double\t");
scanf("%lf",&c);
printf("c = %lf\n",c);    // Output: c = 3.141590

printf("enter the char\t");
scanf(" %c",&d);
printf("d = %c\n",d);    // Output: d = X

printf("enter the string\t");
scanf("%s",e);
printf("e = %s\n",e);    // Output: e = Hello

return 0;
}

```

12 Operators

13.Program on Arithmetic Operators

14.Program on Relational Operators

15.Program on Logical Operators

```

#include <stdio.h>
int main()
{
    // --- 1. Arithmetic Operators ---
    int a1 = 10;
    int b1 = 3;

    printf("Addition = %d\n", a1 + b1);    // Output: Addition = 13
    printf("Subtraction = %d\n", a1 - b1); // Output: Subtraction = 7
    printf("Multiplication = %d\n", a1 * b1); // Output: Multiplication = 30
    printf("Division = %d\n", a1 / b1);    // Output: Division = 3
    printf("Modulus = %d\n", a1 % b1);    // Output: Modulus = 1

    // --- 2. Relational Operators ---
    int a2 = 10;
    int b2 = 20;

    printf("%d\n", a2 == b2);    // Output: 0
    printf("%d\n", a2 != b2);    // Output: 1
    printf("%d\n", a2 > b2);    // Output: 0
    printf("%d\n", a2 < b2);    // Output: 1
    printf("%d\n", a2 >= b2);    // Output: 0
}

```

```

printf("%d\n", a2 <= b2); // Output: 1

// --- 3. Logical Operators ---
int a3 = 10;
int b3 = 20;

printf("%d\n", (a3 < b3) && (a3 > 5)); // Output: 1
printf("%d\n", (a3 > b3) || (b3 > 15)); // Output: 1
printf("%d\n", !(a3 > b3));           // Output: 1

return 0;
}

```

16. Program on Increment/Decrement Operators

```

#include <stdio.h>
int main()
{
    int a = 5;
    int b = 10;

    printf("%d\n", ++a); // Output: 6
    printf("%d\n", a++); // Output: 6
    printf("%d\n", a);   // Output: 7

    printf("%d\n", --b); // Output: 9
    printf("%d\n", b--); // Output: 9
    printf("%d\n", b);   // Output: 8

    return 0;
}

```

17. Expression Evaluation

Precedence	Operators
1	()
2	++ --
3	* / %
4	+ -

```

#include <stdio.h>
int main()
{
    int a = 10, b = 5, c = 2;
    int x = 5;
    int result;

    result = a + b * c;
    printf("%d\n", result);    // Output: 20

    result = (a + b) * c;
    printf("%d\n", result);    // Output: 30

    result = a < 20 && b > 2;
    printf("%d\n", result);    // Output: 1

    result = ++x + 2;
    printf("%d\n", result);    // Output: 8
    return 0;
}

```

18. Basic Technical Program (add of 2 numbers)

19. Fundamentals Programming (swip of 2 numbers)

20. Area of Circle ($3.14 * r * r$)

```

#include <stdio.h>
int main()
{
    float r, a;

    printf("Enter radius: ");    // Output: Enter radius:
    scanf("%f", &r);            // (User enters: 3)

    a = 3.14 * r * r;

    printf("Area = %.2f", a);    // Output: Area = 28.26

    return 0;
}

```

21. Print ASCII Value of a Character (A -> 65 to 90, a -> 97 to 122) difference = 32

```

#include <stdio.h>
int main()
{
    char ch = 'a';
    char ch1 = 'A';

    printf("%d\n", ch);          //97
    printf("%d\n", ch1);         //65
}

```

```

    return 0;
}

```

22. Converting Characters to Upper to Lower Case (32 difference)

```

#include <stdio.h>

int main()
{
    char ch = 'A';
    char ch1 = 'b';

    printf("%c\n",ch + 32);    //a
    printf("%c",ch1 - 32);    //B

    return 0;
}

```

23.Swap Two Numbers

24.Average of 2 no's

```

#include <stdio.h>

int main()
{
    int a = 10;
    int b = 9;
    float avg = (float)(a+b)/2;

    printf("%f",avg);    //9.500000

    return 0;
}

```

25. Fahrenheit to Centigrade (Celsius)

```

#include <stdio.h>
int main()
{
    float fahrenheit, celsius;

    printf("Enter temperature in Fahrenheit: ");
    scanf("%f", &fahrenheit);

    celsius = (fahrenheit - 32) * 5 / 9;

    printf("Temperature in Celsius = %.2f", celsius);

    return 0;
}

```

Output:

Enter temperature in Fahrenheit: 98.6

Temperature in Celsius = 37.00

26. IF Statement

27. Program for Person is eligible for Vote

```
#include <stdio.h>
int main()
{
    int age;

    printf("Enter your age: ");
    scanf("%d", &age);

    if (age >= 18)
    {
        printf("Person is eligible for vote.");
    }
    else
    {
        printf("Person is not eligible for vote.");
    }

    return 0;
}
```

28. Determine the character entered by user

```
#include <stdio.h>

int main()
{
    char ch;

    printf("Enter a character: ");
    scanf(" %c", &ch);

    if ((ch >= 'A' && ch <= 'Z') || (ch >= 'a' && ch <= 'z'))
    {
        printf("Alphabet");
    }
    else if (ch >= '0' && ch <= '9')
    {
        printf("Digit");
    }
    else
    {
        printf("Special Character");
    }

    return 0;
}
```

29. Largest of 2 numbers

30. Given number is even or odd

31.Convert given character is lower to upper & Vice versa

```
#include <stdio.h>

int main()
{
    char ch;

    printf("Enter a character: ");
    scanf(" %c", &ch);

    if (ch >= 'A' && ch <= 'Z')
    {
        ch = ch + 32;
        printf("Lowercase = %c", ch);
    }
    else if (ch >= 'a' && ch <= 'z')
    {
        ch = ch - 32;
        printf("Uppercase = %c", ch);
    }
    else
    {
        printf("Not an alphabet.");
    }

    return 0;
}
```

32.Switch Statement

```
#include <stdio.h>

int main()
{
    int a, b;
    char op;

    printf("Enter first number: ");
    scanf("%d", &a);

    printf("Enter operator (+, -, *, /): ");
    scanf(" %c", &op);

    printf("Enter second number: ");
    scanf("%d", &b);

    switch(op)
    {
        case '+':
            printf("Result = %d", a + b);
            break;

        case '-':
```

```

        printf("Result = %d", a - b);
        break;

    case '*':
        printf("Result = %d", a * b);
        break;

    case '/':
        printf("Result = %d", a / b);
        break;

    default:
        printf("Invalid Operator");
    }

    return 0;
}

```

Output:

```

Enter first number: 5
Enter operator (+, -, *, /): +
Enter second number: 8
Result = 13

```

33. Demonstrate switch statement without break

```
#include <stdio.h>
```

```

int main()
{
    int choice;

    printf("Enter a number (1-3): ");
    scanf("%d", &choice);

    switch(choice)
    {
        case 1:
            printf("One\n");

        case 2:
            printf("Two\n");

        case 3:
            printf("Three\n");

        default:
            printf("Invalid\n");
    }

    return 0;
}

```

```

1
ONE
TWO
THREE
INVALID

3
THREE
INVALID

```

34. Determine Whether Entered Character is Vowel or Not

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    char ch;
```

Enter a character: A

Vowel

```
    printf("Enter a character: ");
```

```
    scanf(" %c", &ch);
```

Enter a character: z

Not a Vowel

```
    switch(ch)
```

```
    {
```

```
        case 'a':
```

```
        case 'e':
```

```
        case 'i':
```

```
        case 'o':
```

```
        case 'u':
```

```
        case 'A':
```

```
        case 'E':
```

```
        case 'I':
```

```
        case 'O':
```

```
        case 'U':
```

```
            printf("Vowel");
```

```
            break;
```

```
        default:
```

```
            printf("Not a Vowel");
```

```
    }
```

```
    return 0;
```

```
}
```

35. Enter a Number from 1 to 7 and Display Corresponding Day of Week

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int day;
```

```
    printf("Enter a number (1-7): ");
```

```
    scanf("%d", &day);
```

```
    switch(day)
```

```
    {
```

```
        case 1:
```

```
            printf("Sunday");
```

```
            break;
```

```
        case 2:
```

```
            printf("Monday");
```

```
            break;
```

```
        case 3:
```

```
            printf("Tuesday");
```

```
            break;
```

```
        case 4:
```

```
            printf("Wednesday");
```

```
            break;
```

Enter a number (1-7): 5

Thursday

Enter a number (1-7): 9

Invalid Day

```
    case 5:
        printf("Thursday");
        break;
    case 6:
        printf("Friday");
        break;
    case 7:
        printf("Saturday");
        break;
    default:
        printf("Invalid Day");
}

return 0;
}
```